Propagating Protein-Protein Interaction Network Support into GRNsight 7.0, a Web Application for Visualizing Gene Regulatory Network Models

Ngoc K. Tran[1], Cecilia J. Zaragoza[1], A'Kaia Phelps[1], Kam D. Dahlquist[2], John David N. Dionisio[1]

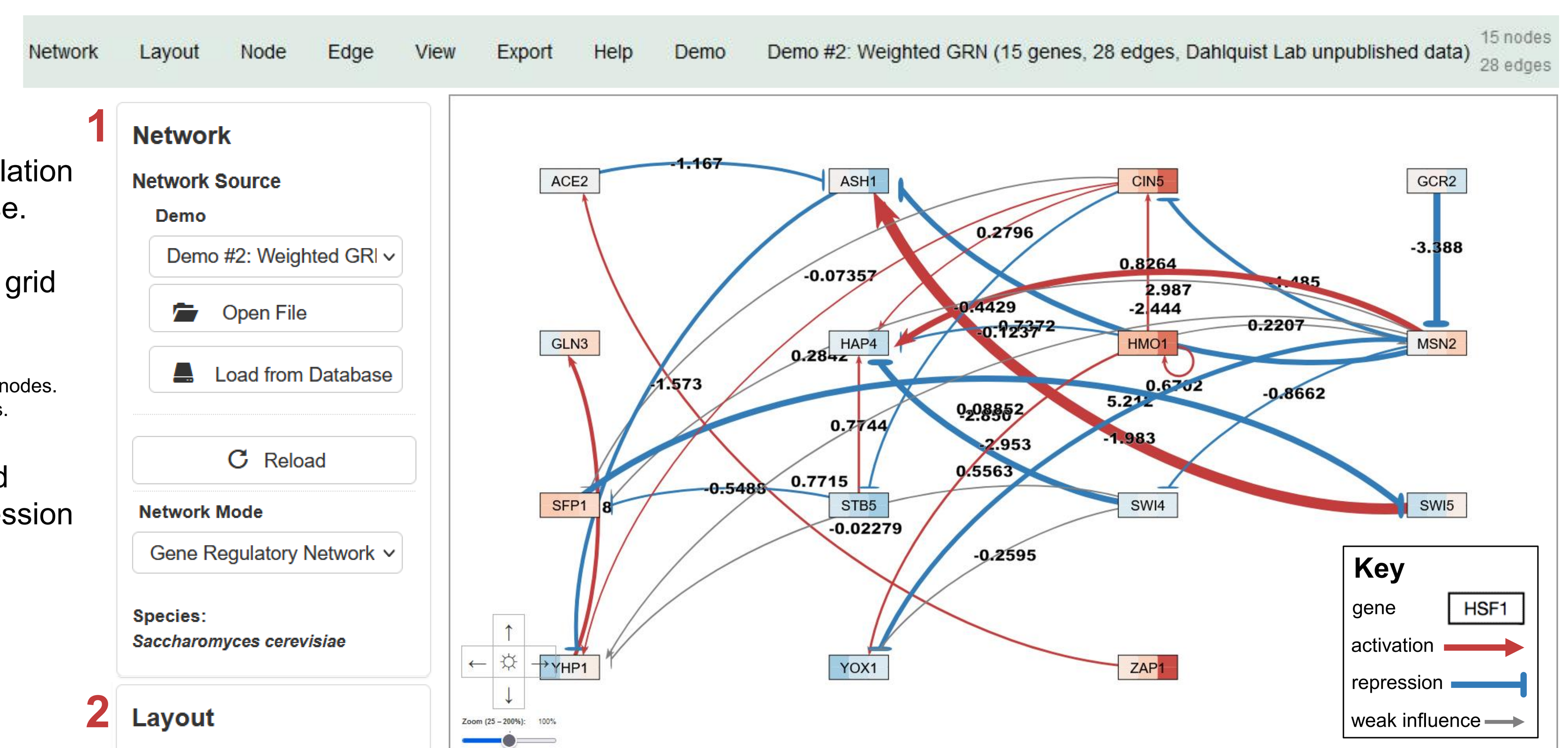
Departments of [1]Computer Science and [2]Biology, Loyola Marymount University, 1 LMU Drive, Los Angeles, CA 90045 USA

http://dondi.github.io/GRNsight/

Loyola Marymount University

GRNsight, automatically lays out a gene regulatory network (GRN) graph, coloring the edges based on regulatory weights, and the nodes based on experimental or simulated expression data

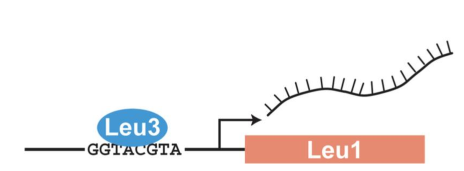
1. **Network**
   - Demo files are provided.
   - Can import and export Excel, SIF, or GraphML files.
   - Can construct a network from SGD regulation or PPI data stored in a backend database.
2. **Layout**
   - Users can select a force graph layout or grid layout.
   - Force graph parameter sliders:
     - Link distance determines the minimum distance between nodes.
     - Nodes have a charge which repels or attracts other nodes.
3. **Node**
   - Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
   - Datasets available:
     - Apweiler et al. 2012, GSE33098
     - Barreto et al. 2012, GSE24712
     - Dahlquist et al. 2018, GSE83656
     - Kitagawa et al. 2002, GSE9336
     - Thorsen et al. 2007, GSE6068
4. **Edge**
   - Edge coloring can be toggled on and off.
   - Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover.
   - Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
   - Changing the gray threshold deemphasizes weaker edges in the visualization.
   - Presenting gray edges as dashed lines further deemphasizes weak edges.
5. **View**
   - Multiple viewport sizes available.
   - Graph bounding box can be separated from viewport.
   - Zooming and scrolling available.



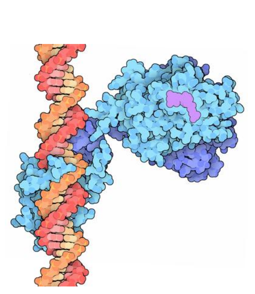
*Some new features are only available in GRNsight v7.0.13 beta.

GRNsight can now be used to visualize protein-protein physical interaction (PPI) networks

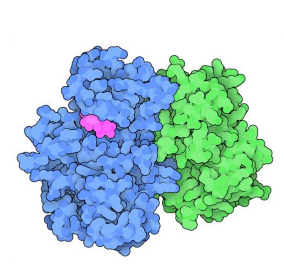
- **A gene regulatory network consists of genes, transcription factors, and the regulatory connections between them which govern the level of expression of mRNA and protein from genes.**
  - Nodes represent both the regulatory transcription factor protein and the gene that encodes it.
  - An arrow represents a regulatory transcription factor either activating (arrowhead, red color) or repressing (blunt end, blue color) the gene for another regulatory transcription factor.
- **A PPI is when two proteins bind to each other.**
  - Each node represents a protein, and each edge represents their binding interaction.
  - Unlike GRNs, PPI graphs have undirected edges of uniform thickness and black color and do not display expression data on the nodes.



Transcription Factors Control Gene Expression by Binding to Regulatory DNA Sequences
- **Activators increase gene expression**
- **Repressors decrease gene expression**
- **Transcription factors are themselves proteins that are encoded by genes**



Regulatory Transcription Factor Bound to DNA
Image from: https://pdb101.rcsb.org/motm/155



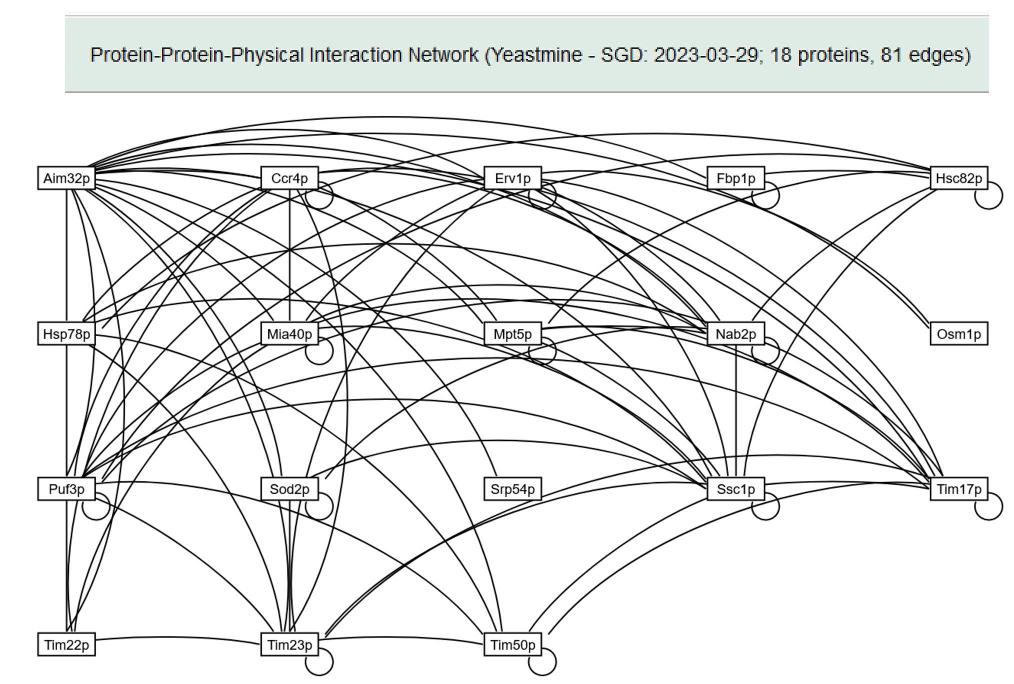
Protein-Protein Interaction
Image from: https://pdb101.rcsb.org/motm/236

- **Loading a PPI Network: 3 Different Ways**
  - Choose the new PPI Demo file
    - Navigate to the Network sidebar or the Demo tab of the navigation bar and choose Demo #5, or load from database specific proteins or graph to generate a network.
  - Upload a PPI (with the column A1 altered)
    - Can upload an Excel file with PPI data, but the cell A1 must be altered to say "cols protein1/ rows protein2".

cols protein1/ rows protein2	Aim32p	Ccr4p	Erv1p
Aim32p	0	0	0
Ccr4p	1	1	0
Erv1p	0	0	1

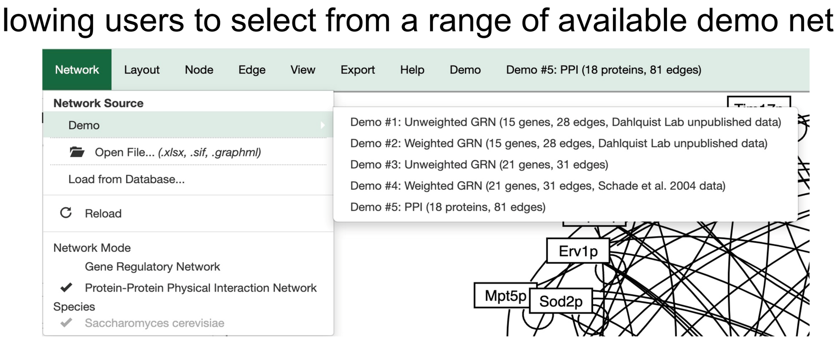
- Upload a SIF (simple interaction format)
  - Upload a SIF with the format '**Protein1 pd Protein2**'.
- GraphML not available for PPI
  - We currently do not support upload with GraphML.

As a result of the incorporation of PPI, GRNsight required further enhancements



- Example of an 18-protein, 81-edge network representing proteins that interact with Aim32p.
- Aim32p is a dual-localized 2Fe-2S mitochondrial protein that functions in redox quality control.

1. **Distinguishing between Directed vs. Undirected Edges**
   - Incorporating visualization techniques such as arrowheads for directed edges and absence of arrowheads for undirected edges, facilitates the accurate representation and interpretation of regulatory and physical interactions within these networks.
     - The Directed Graph represent connections in GRN (gene-regulatory-network)
     - Demonstrating different types of network types:
       - In the unweighted GRN SIF file, interactions represent regulatory relationships between genes or transcription factors. Each line indicates a regulatory interaction, with the format '**Regulator pd Target**'.
     - The Undirected Graph represent interactions of PPI (protein-protein physical interactions)
2. **PPI Demo Graph Reference**
   - The dropdown menu serves as an interactive feature in GRNsight, allowing users to select from a range of available demo networks.



3. **SIF File (PPI and GRN)**
   - Demonstrating different types of network types:
     - In the unweighted GRN SIF file, interactions represent regulatory relationships between genes or transcription factors. Each line indicates a regulatory interaction, with the format '**Regulator pd Target**'.
     - In the PPI SIF file, interactions represent physical associations between proteins. Each line denotes a physical interaction between proteins, with the format '**Protein pp Protein**'
     - Can be viewed and exported with Excel, VSCode, Word, and more.

GRN SIF

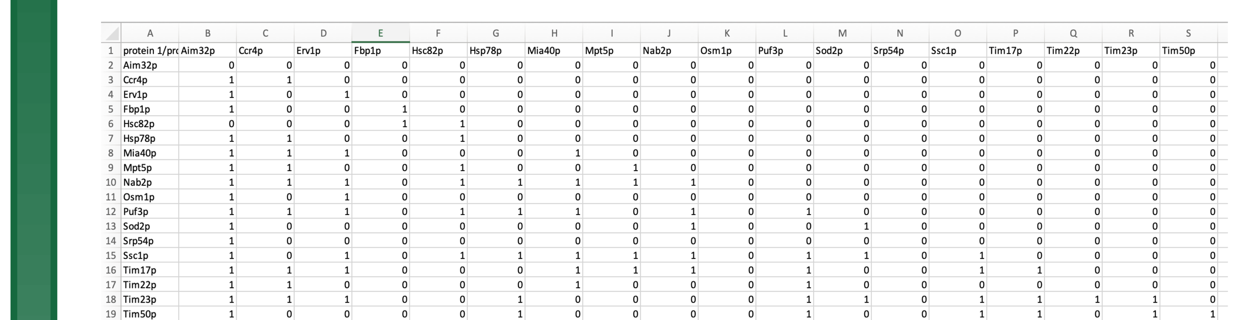
```
1  ACE2  pd  ASH1
2  ASH1  pd  YHP1
3  CIN5  pd  HAP4
4  CIN5  pd  SFP1
5  CIN5  pd  STB5
6  CIN5  pd  YHP1
7  GCR2  pd  MSN2
8  GLN3
9  HAP4
10 HMO1  pd  CIN5
```

PPI SIF

```
1  Aim32p  pp  Ccr4p
2  Aim32p  pp  Erv1p
3  Aim32p  pp  Fbp1p
4  Aim32p  pp  Hsp78p
5  Aim32p  pp  Mia40p
6  Aim32p  pp  Mpt5p
7  Aim32p  pp  Nab2p
8  Aim32p  pp  Osm1p
9  Aim32p  pp  Puf3p
10 Aim32p  pp  Sod2p
```

Network Mode Detection

1. **Network Mode Imports**
   - SIF files are detected based on either "pp" or "pd" indicator, refer to last image. GraphML files are always associated with GRN. (refer to image in the last section)
   - Excel files are identified based on the naming convention in cell A1, with "cols regulator/ rows target" for GRN and "cols protein1/ rows protein2" for PPI.



Ongoing Development

1. **Enabling Controls of D-pad, Zoom Slider, and Cursor Grab When Restrict Graph to Viewport**
   - Previously when the graph was restricted to the viewport the D-pad, Zoom Slider, and Cursor Grab controls were disabled. The user is now allowed access to those controls to navigate around the viewport and move the graph.
   - The goal is to allow users to utilize the D-pad, Zoom Slider, and Cursor Grab controls while ensuring that the nodes remain inside the bounds of the viewport. The nodes will be clamped inside a bounding box, which will be outlined in gray, that will designate the maximum space the nodes can occupy.

Future Directions

- Automate the process of toggling between a particular PPI network and the GRN that regulates the genes encoding the proteins in the PPI network, as shown above.
- Update the backend GRNsight GRN and PPI databases with 2024 data from SGD.
- Prioritize bug resolution to enhance stability and reliability of the platform.
- Focus on user interface enhancements to improve usability and user experience.
- Improve documentation to provide clear and comprehensive guidance for users.

Availability

- GRNsight is free and open to all users at http://dondi.github.io/GRNsight/, and there is no login requirement.
- GRNsight code is available under the open source BSD license at our GitHub repository at https://github.com/dondi/GRNsight.
- Some new GRNsight features are only available in GRNsight v7.0.13 beta.

Acknowledgments

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